

Chapter 16: Genetics — Revision Notes

Williams Obstetrics 26e, pp. 789–848 of the PDF. High-yield numbers are in **bold**. The seven tables are transcribed from the PDF images (they are missing from the extracted chapter text). 13 of the 14 figures are included (Fig 16-11, a FISH probe-making schematic, is omitted).

Big picture: Genetic disease is common: **2–3%** of newborns have a recognized structural abnormality, another **3%** are diagnosed by age 5, and **8–10%** more have a functional or developmental abnormality before adulthood. Chromosomal, mendelian and nonmendelian conditions are covered here; prenatal screening/diagnosis is in Ch 17.

1. Genomics in Obstetrics

- **Gene** = specific DNA sequence coding for a protein; **genome** = all genes of an organism; **genomics** = how genes function and interact.
- Human genome: **3 billion base pairs**, **>20,000 genes**. **>99%** of DNA is identical between people.
- **Exons (coding)** = **only 1.5%** of the genome; **exome** = all exons. Introns = 24%; the rest is intergenic.
- Genetic code varies once every **200–500 bp**, usually as a **single-nucleotide polymorphism (SNP)**.
- Databases (NCBI): **GeneReviews** (>800 conditions; diagnostic criteria, management), **Genetic Testing Registry** (~80,000 tests), **MedlinePlus Genetics** (lay public, >1300 conditions).

2. Chromosomal Abnormalities

- Account for **>50% of 1st-trimester miscarriages**, **~20% of 2nd-trimester losses**, and **6–8% of stillbirths** and early-childhood deaths.
- Found in **0.4% of recognized pregnancies**; most are aneuploidies.
 - **Trisomy 21 >50%**, **trisomy 18 ~15%**, **trisomy 13 ~5%**.
- Fetus with a structural anomaly and a normal karyotype → CMA finds a **copy number variant in ~6.5%**.

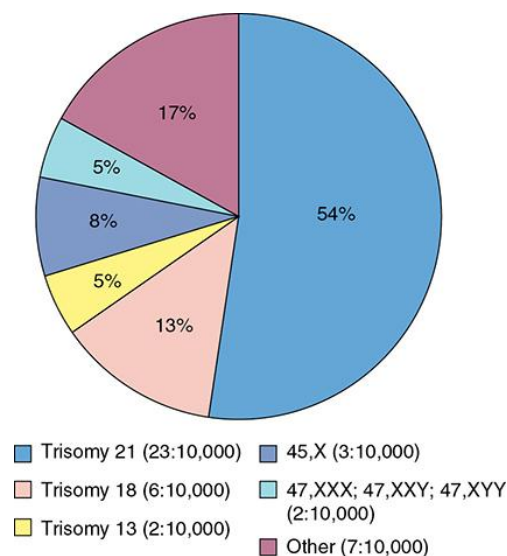


Figure 16-1. EUROCAT: trisomy 21 is 54% of chromosomal abnormalities (23:10,000), T18 13%, T13 5%, 45,X 8%, other sex aneuploidies 5%.

Standard nomenclature (ISCN 2020)

- **p** = short ("petit") arm; **q** = long arm; separated by the **centromere**.

- Karyotype order: **total chromosome number** (= number of centromeres) → **sex chromosomes** → structural variation.
- Abbreviations: **del** (deletion), **dup**, **inv** (inversion), **t** (translocation), **der** (derivative), **r** (ring), **i** (isochromosome).
- FISH: **ish** = metaphase cells; **nuc ish** = interphase nuclei. Normal: probe region + probe name + number of signals (e.g., HIRAx2). Deletion: "del" before the region and a minus sign after the probe (HIRA-).
- CMA: starts with **arr**, then genome build (e.g., **GRCh38**), chromosome, arm, bands and **base-pair coordinates** (gives exact size and location). Same format whether pathogenic or of uncertain significance.

Table 16-1. Examples of Karyotype Designations Using the 2020 International System for Human Cytogenomic Nomenclature

Karyotype	Description
46,XX	Normal female chromosome constitution
47,XY,+21	Male with trisomy 21
47,XX,+21/46,XX	Female who is a mosaic of trisomy 21 cells and cells with normal constitution
46,XY,del(4)(p14)	Male with terminal deletion (del) of the short arm of chromosome 4 at band p14
46,XX,dup(5)(p14p15.3)	Female with duplication (dup) of the short arm of chromosome 5 from band p14 to band p15.3
45,XY,der(13;14)(q10;q10)	Male with balanced robertsonian translocation (der) of the long arms of chromosomes 13 and 14 — the karyotype now has one normal 13, one normal 14 and the translocation chromosome, reducing the normal 46 complement to 45
46,XX,t(11;22)(q23;q11.2)	Female with a balanced reciprocal translocation (t) between chromosomes 11 and 22, with breakpoints at 11q23 and 22q11.2
46,XY,inv(3)(p21q13)	Male with inversion (inv) of chromosome 3 from p21 to q13 — a pericentric inversion because it includes the centromere
46,X,r(X)(p22.1q27)	Female with one normal X and one ring (r) X chromosome, with the regions distal to p22.1 and q27 deleted from the ring
46,X,i(X)(q10)	Female with one normal X and an isochromosome (i) of the long arm of the other X
ish 22q11.2(HIRAx2)	FISH of metaphase cells using a probe for the HIRA locus of the 22q11.2 region, with 2 signals (no evidence of microdeletion)
ish del(22)(q11.2q11.2)(HIRA-)	FISH of metaphase cells using a probe for the HIRA locus of 22q11.2, with only one signal, consistent with the microdeletion
arr[GRCh38] 18p11.32q23(102328_79093443)x3	Microarray (arr), build GRCh38, single copy gain of chromosome 18 from p11.32 to q23 (essentially the entire chromosome), consistent with trisomy 18
arr[GRCh38] 4q32.2q35.1(163146681_183022312)x1	Microarray, build GRCh38, copy loss on the long arm of chromosome 4 at bands q32.2 through q35.1 (19.9 Mb)
arr[GRCh38] 15q11.2q26(23123715_101888908)x2 hnz	SNP microarray, build GRCh38, homozygosity for the entire long arm of chromosome 15

FISH = fluorescence in situ hybridization; GRCh38 = Genome Reference Consortium human build 38; HIRA = histone cell cycle regulator; SNP = single-nucleotide polymorphism.

3. Abnormalities of Chromosome Number

- **Aneuploidy** = extra (trisomy) or missing (monosomy) chromosome. **Polyploidy** = abnormal number of whole haploid sets (e.g., **triploidy = 69**).

Autosomal trisomies — general

- Usually from **nondisjunction**: chromosomes (1) fail to pair, (2) pair but separate prematurely, or (3) fail to separate.
- Any pair can missegregate, but only **21, 18 and 13** commonly reach term.
- Risk rises steeply with maternal age, **especially after 35**. Oocytes are arrested in **midprophase of meiosis I** from birth until ovulation.
- **10–20% of oocytes** are aneuploid vs **3–4% of sperm**.
- **Recurrence after any autosomal trisomy $\approx 1\%$** (until age-related risk exceeds this).
- **Parental karyotype is NOT indicated** unless caused by an unbalanced translocation or other structural rearrangement.

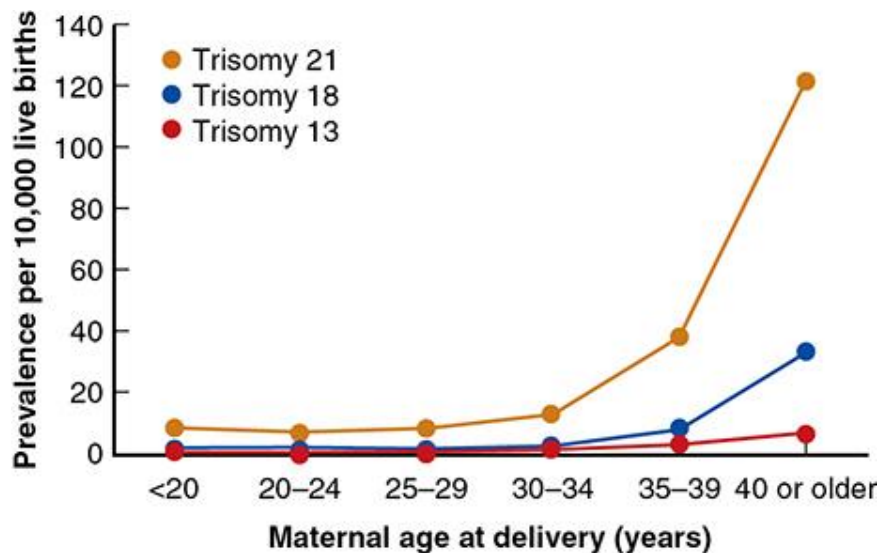


Figure 16-2. Trisomy prevalence by maternal age: flat until the early 30s, then rises steeply (T21 $\sim 120/10,000$ at ≥ 40).

Trisomy 21 — Down syndrome

- **Causes**: trisomy 21 **95%** (nondisjunction in **meiosis I** in **$\sim 75\%$** , meiosis II in the rest); **robertsonian translocation 3–4%**; isochromosome 21 or mosaicism **1–2%**.
- **Most common nonlethal trisomy**: **~ 1 in 450 pregnancies**; **1 in 740 live births** (US). Fetal death >20 wk $\sim 5\%$.
- Proportion of pregnancies affected has risen **$\sim 33\%$** over 4 decades (older mothers).
- **Sonographic abnormality in 50–75%** (major anomalies + minor markers); a major malformation is seen in the 2nd trimester in **$\sim 1/3$** .
- **Cardiac defects $\sim 50\%$** — especially **endocardial cushion defects and VSD**.
- **GI defects 6–12%**: esophageal atresia, **duodenal atresia**, Hirschsprung disease.
- Other problems:

Problem	Frequency
Hearing loss	75%
Early-onset Alzheimer disease (adults)	70%
Obstructive sleep apnea	60%
Refractive errors	50%
Cataracts	15%
Thyroid disease	15%
Transient myeloproliferative disorder (newborns)	10%
Leukemia	$\uparrow$ incidence

- Women with DS are fertile — **$\sim 1/3$ of offspring have DS**. Males are almost always **sterile**.

- **IQ 35–70**; social skills higher than IQ predicts.
- **Features:** brachycephaly, epicanthal folds, up-slanting palpebral fissures, **Brushfield spots**, flat nasal bridge, hypotonia, loose nuchal skin, **sandal gap**, short fingers, **single palmar crease**, **5th-finger clinodactyly** (hypoplastic middle phalanx).
- **Survival:** ~**95%** survive year 1; 10-yr survival **≥90%** (**99%** without major malformations). Life expectancy **55–60 yr**.

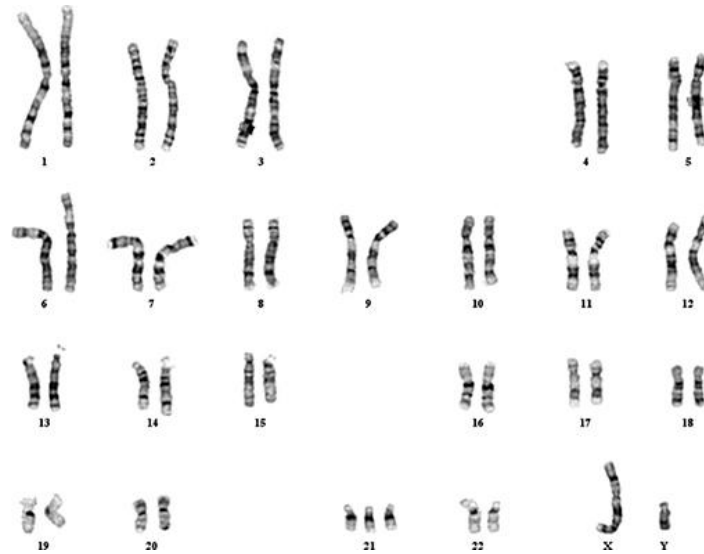


Figure 16-3. G-banded male karyotype 47,XY,+21: three copies of chromosome 21.

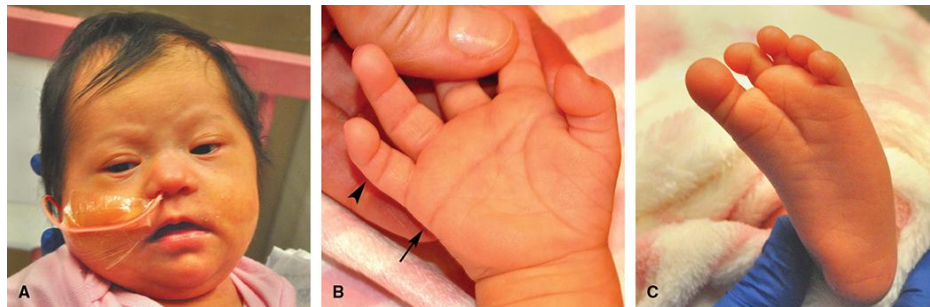


Figure 16-4. Down syndrome newborn: (A) epicanthal folds, flat nasal bridge; (B) single palmar crease (arrow) and 5th-digit clinodactyly (arrowhead); (C) sandal gap.

Trisomy 18 — Edwards syndrome

- ~**1 in 2000 recognized pregnancies**; live births only **1 in 6600–10,000** (high in-utero lethality + termination).
- **Uncommonly** from a rearrangement (chromosome 18 is not acrocentric — unlike 21 and 13).
- **Heart defects >90%** (especially VSD). Also: cerebellar vermian agenesis, myelomeningocele, diaphragmatic hernia, **omphalocele**, imperforate anus, horseshoe kidney.
- Cranial/limb signs: "**strawberry**" skull, wide cavum septum pellucidum, **choroid plexus cysts**, micrognathia, **clenched hands with overlapping digits**, **radial aplasia** with hyperflexed wrists, **rockerbottom or clubbed feet**.
- ⚠️ **Choroid plexus cysts raise T18 risk only with other risk factors** (structural anomaly or abnormal screen).
- 3rd trimester: **FGR**, mean birthweight **<2500 g**. Undiagnosed → emergency cesarean for "fetal distress" in **~50%** → discuss mode of delivery/FHR management in advance; offer **perinatal palliative care**.
- **Median neonatal survival 8 days; 5-yr survival 12%**. Cardiac surgery mortality **3–5x** higher.

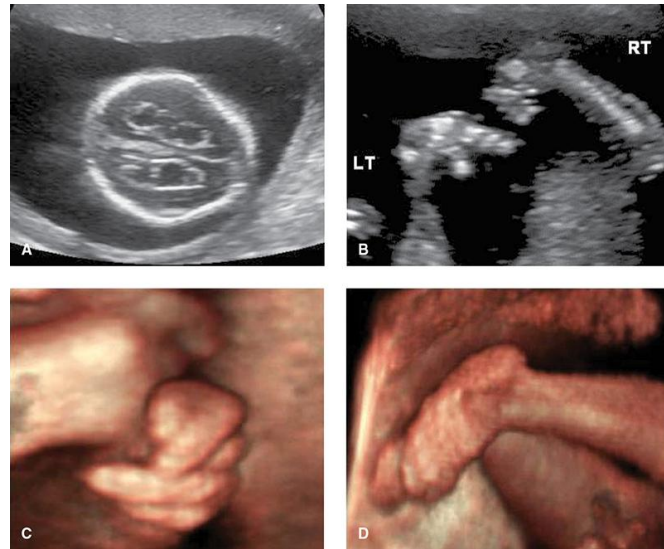


Figure 16-5. Trisomy 18: (A) choroid plexus cysts and strawberry skull; (B) radial clubhand; (C) clenched fist with overlapping digits; (D) rockerbottom foot.

Trisomy 13 — Patau syndrome

- **~1 in 5000 recognized pregnancies**; live births **1 in 12,000–18,000**.
- **≥80% are trisomy 13**; nearly all the rest are **robertsonian translocations**, most commonly **der(13;14)(q10;q10)** (carried by **~1 in 1300** normal people; **<2%** of carriers have a liveborn with Patau).
- Brain, heart, kidneys, extremities most affected:
 - **Holoprosencephaly (usually alobar) in ⅔** — with hypotelorism/cyclopia, microphthalmia, single nostril to proboscis.
 - **Cardiac defects up to 90%**.
 - Cephalocele, microcephaly, omphalocele, **cystic renal dysplasia**, **polydactyly**, rockerbottom feet, **aplasia cutis**, cleft lip-palate (may be median), **bilateral echogenic intracardiac foci**.
- **Cephalocele + cystic kidneys + polydactyly** → differential = **T13 vs Meckel–Gruber syndrome** (autosomal recessive).
- More lethal than T18: **median survival 5 days; 5-yr survival <10%**.
- **⚠ Maternal risk**: continuing pregnancies have **≥25% hypertensive complications; preeclampsia with severe features ↑ >10-fold**, often **<32 wk**. Chromosome 13 carries the **sFlt-1** gene (overexpressed by T13 placentas).

Other trisomies

- Live births extremely rare. **Trisomy 22**: severe FGR, microcephaly, midface hypoplasia, cardiac/GU anomalies.
- **Trisomy 16 = most common trisomy in 1st-trimester losses (16%)**; T21 and T22 also common. **Trisomy 1 never reported**.

Monosomy

- Missing material is worse than extra: almost all monosomies are lost **before implantation**.
- **Only exception: 45,X**. Monosomy has **no maternal-age association**.

Polyploidy

- **~20% of spontaneous abortions**; rare later.
- **Triploidy (69 chromosomes)**, **~1 in 5000** pregnancies, lethal; multiple anomalies (brain, heart, kidneys, limbs). Phenotype depends on parent of origin (**imprinting**):

	Diandric (type I)	Digynic (type II)
Extra set	Paternal	Maternal
Mechanism	2 sperm (dispermy) or 1 diploid sperm	Egg fails meiosis I or II
Placenta	Partial mole (Ch 13)	No molar change
Fetus	—	Asymmetric FGR
Frequency	Majority of conceptions, but high early loss	¼ of triploidies seen after 1st trimester

- **Recurrence 1–1.5%** if the triploid fetus survived past the 1st trimester → offer prenatal diagnosis.
- **Tetraploidy** (92,XXXX or 92,XXYY): postzygotic failure of early cleavage; always lethal; minimal recurrence.

Sex chromosome abnormalities

45,X — Turner syndrome

- **~1 in 2500 girls**; missing X is **paternal in 80%**.
- **Only monosomy compatible with life**, yet the **most common aneuploidy in 1st-trimester losses (20%)**. **~98%** abort early.
- Survivors to 2nd trimester: large **septated cystic hygromas** ± hydrops (**very poor prognosis** with hydrops). **<1%** of Turner pregnancies → live birth.
- Liveborns: only **½ are 45,X**; **~¼ mosaic** (45,X/46,XX or 45,X/46,XY); **15% isochromosome X**.
- **Left-sided cardiac defects 30–50%**: **coarctation**, hypoplastic left heart, **bicuspid aortic valve**. Also **horseshoe kidney**, hypothyroidism.
- Short stature, broad chest with wide-spaced nipples, **lymphedema** of hands/feet, **webbed neck** (from cystic hygroma).
- IQ usually normal (visual-spatial, nonverbal and social-cue problems).
- **GH** in childhood; **>90% have ovarian dysgenesis** → estrogen at puberty.
- **⚠ Y-chromosome mosaicism → germ cell tumor risk → prophylactic bilateral gonadectomy.**

47,XXX

- **~1 in 1000** female newborns; extra X **maternal in >90%**; weak maternal-age link; more diagnoses now via cfDNA.
- No characteristic appearance; ↑ GU problems and seizures. Tall, hypertelorism, epicanthal folds, kyphoscoliosis, clinodactyly, hypotonia.
- **>½ learning disability**, **½ ADD**; low-average cognition. Normal puberty; POI reported.
- Only **~10%** ascertained without prenatal diagnosis.
- 48,XXXX / 49,XXXXX: abnormalities at birth + intellectual disability. **IQ falls with each extra X.**

47,XXY — Klinefelter syndrome

- **Most common sex chromosome abnormality: ~1 in 700 males**. Extra X maternal or paternal equally; weak parental-age link.
- Normal newborn appearance; tall; **gonadal dysgenesis**, poor virilization → **testosterone**; gynecomastia.
- IQ average/low-average, delayed language. **Earlier (childhood) hormone therapy** → better working memory/executive function, less anxiety.

47,YYY

- **~1 in 1000 males**. Tall; **macrocephaly ⅓**, **hypotonia ~⅓**, tremors; hypertelorism and clinodactyly in >½.
- **Normal puberty and fertility**. ↑ language impairment, ADD, developmental delay, **autism spectrum disorder**.
- 48,YYYY, 48,XXYY, 49,XXXYY → more anomalies and intellectual disability.

4. Abnormalities of Chromosome Structure

- Types: deletions, duplications, translocations, isochromosomes, inversions, rings, mosaicism.

- Two key questions: **(1) what phenotype is expected? (2) are the parents carriers (is parental karyotyping indicated)?**

Deletions and duplications

- From **malalignment/mismatching during pairing** of homologues in meiosis → deletion in one chromosome, duplication in the other.
- **Offer parental karyotyping** — a parental balanced translocation markedly ↑ recurrence.
- Karyotype-visible deletions: **~1 in 7000 births** (e.g., **del 5p = cri du chat**).

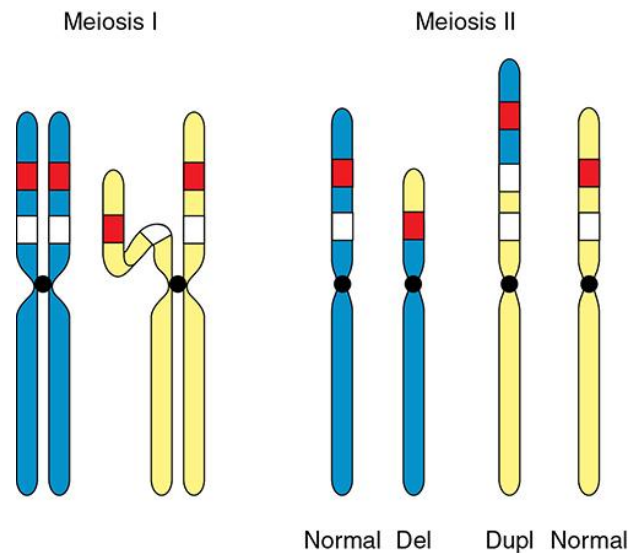


Figure 16-6. Mismatched pairing in meiosis I produces one deleted (Del) and one duplicated (Dupl) chromosome.

Microdeletions and microduplications

- **<3–5 million bp** → too small for a karyotype; found by **CMA** (called a **copy number variant**).
- Multigene region → **contiguous gene syndrome** (unrelated but serious features). A microduplication may involve the same region as a known microdeletion syndrome.
- Suspected clinically → confirm by **CMA or FISH**.

Table 16-2. Selected Microdeletion Syndromes

Syndrome	Prevalence	Location	Features
Alagille	1:70,000	20p12.2	Cholestasis (paucity of intrahepatic bile ducts), cardiac disease, skeletal disease, ocular abnormalities, dysmorphic facies
Angelman	1:12,000 to 1:20,000	15q11.2–q13 (maternal genes)	Dysmorphic facies — "happy puppet" appearance, intellectual disability, ataxia, hypotonia, seizures
Cri-du-chat	1:20,000 to 1:50,000	5p15.2–15.3	Abnormal laryngeal development with "cat-like" cry, hypotonia, intellectual disability
Kallmann syndrome	1:30,000 males	Xp22.3	Hypogonadotropic hypogonadism, anosmia
Langer-Giedion	Rare	8q23.3	Trichorhinophalangeal syndrome, dysmorphic facies, skeletal abnormalities, sparse hair
Miller-Dieker	Rare	17p13.3	Neuronal migration abnormalities with lissencephaly and microcephaly (profound impairment), dysmorphic facies
Prader-Willi	1:10,000 to 1:30,000	15q11.2–q13 (paternal genes)	Obesity, hypotonia, hypogonadotropic hypogonadism, small hands and feet, intellectual disability
Retinoblastoma	1:280,000	13q14.2	Retinoblastoma, retinoma (benign neoplasm), non-retinal (second primary) tumors
Rubenstein-Taybi	1:100,000 to 1:125,000	16p13.3	Dysmorphic facies, broad thumbs and toes, intellectual disability, increased tumor risk
Smith-Magenis	1:15,000 to 1:25,000	17p11.2	Dysmorphic facies, speech delay, hearing loss, sleep disturbances, self-destructive behaviors, intellectual disability
Velocardiofacial syndrome	1:3000 to 1:6000	22q11.2	Conotruncal cardiac defects, cleft palate, velopharyngeal incompetence, thymic and parathyroid abnormalities, intellectual disability
WAGR	1:500,000	11p13	W ilms tumor, a niridia, g enitourinary anomalies, intellectual disability (mental retardation)
Williams-Beuren	1:7500 to 1:10,000	7q11.23	Dysmorphic facies, dental malformation, aortic and peripheral pulmonary artery stenosis, intellectual disability
Wolf-Hirschhorn	1:20,000 to 1:50,000	4p16.3	Dysmorphic ("Greek helmet") facies, delayed growth and development, cleft lip/palate, coloboma, cardiac septal defects
X-linked ichthyosis	1:6000	Xp22.3	Steroid sulfatase deficiency, corneal opacities

Prevalence reflects live births.

22q11.2 microdeletion syndrome (DiGeorge / Shprintzen / velocardiofacial)

- **Most common microdeletion: 1 in 3000–6000 births.**
- **Autosomal dominant** (50% to offspring), but **>90% de novo**. Full deletion = **3 Mb, 40 genes, 180 features**; variable even within families → offer testing to both parents after prenatal diagnosis.
- **Conotruncal cardiac anomaly ~75%**: tetralogy of Fallot, truncus arteriosus, interrupted aortic arch, VSD.
- **Immune deficiency (T-cell lymphopenia) 75%; velopharyngeal insufficiency/cleft palate >70%.**
- Face: short palpebral fissures, bulbous nasal tip, micrognathia, short philtrum, small/posteriorly rotated ears.
- Also **hypocalcemia**, renal anomalies, esophageal dysmotility, hearing loss, learning disability, autism, psychiatric illness (**schizophrenia**).

Memory hook: CATCH-22 — **C**ardiac, **A**bnormal facies, **T**hymic hypoplasia (T-cell deficit), **C**left palate, **H**ypocalcemia, chromosome **22**.

Translocations

Rearranged chromosomes = **derivative (der)**.

Reciprocal translocations

- Breaks in two chromosomes with exchange of fragments; **balanced** if nothing is gained or lost. **~1 in 600 births**.
- Balanced carriers are usually normal, but **~6%** have abnormalities from gene repositioning; **CMA finds hidden imbalance in up to 20%** of "balanced" carriers.
- Risk of a **liveborn with an unbalanced translocation**:
 - **5–30%** if ascertained after an abnormal child
 - **~5%** if ascertained otherwise (e.g., infertility) — gametes are so abnormal that conceptions are nonviable

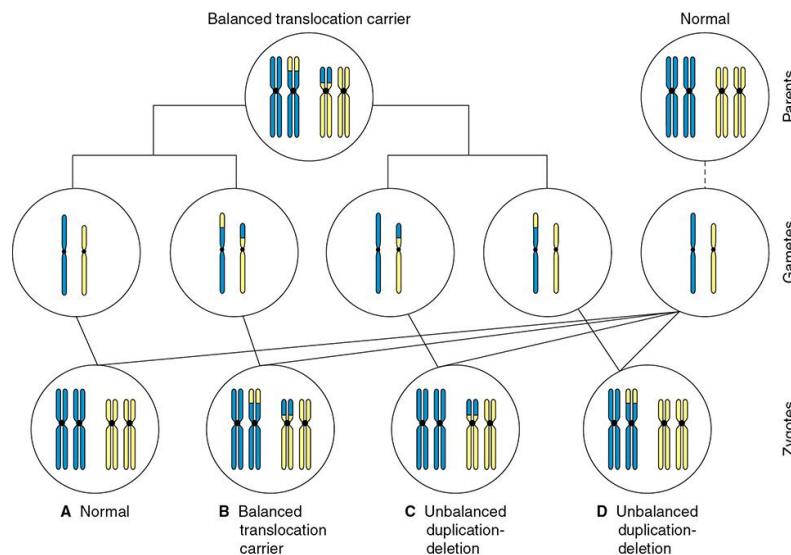


Figure 16-7. Balanced translocation carrier × normal partner: normal (A), balanced carrier (B) or unbalanced duplication–deletion (C, D) offspring.

Robertsonian translocations

- Only **acrocentric chromosomes: 13, 14, 15, 21, 22** (p arms carry redundant rRNA genes).
- The **q arms fuse at one centromere**; the p arms are lost → carrier is normal but has **45 chromosomes**.
- **Homologous** fusion (e.g., 21;21) → **only unbalanced gametes** (all trisomic or monosomic conceptions).
Nonhomologous → **4 of 6** possible gametes abnormal.
- **~1 in 1000** individuals.
- **Risk of abnormal offspring: ~15% if the mother is the carrier, ~2% if the father.**
- **<5%** of couples with recurrent pregnancy loss; not a major miscarriage cause.
- Translocation trisomy in a fetus/child → **karyotype both parents**; if neither carries it, recurrence is extremely low.
- **der(13;14)(q10;q10)** is the most common; causes **up to 20%** of Patau syndrome.

Isochromosomes

- **Two p arms or two q arms fuse** (transverse centromere split in meiosis II/mitosis, or a meiotic error in a robertsonian chromosome).
- **Acrocentric** isochromosome (two q arms) behaves like a **homologous robertsonian** translocation: normal carrier, **only unbalanced gametes**.
- **Nonacrocentric**: trisomy for one arm, monosomy for the other → carrier usually **abnormal**.
- Most common: **i(Xq)** → **15% of Turner syndrome**.

Inversions

Two breaks in one chromosome with the segment inverted; nothing gained or lost, but gene function may change.

	Pericentric	Paracentric
Breaks	Both p and q arms — includes the centromere	One arm — excludes the centromere
Gametes	Abnormal gametes → duplication/deletion offspring	Normal balanced, or too abnormal to be fertilized
Risk of abnormal offspring	5–10% if found after an abnormal child; 1–3% if found for another reason	Extremely low (infertility/early loss may be a problem)
Note	inv(9)(p11q12) = normal variant (~1% of population)	—

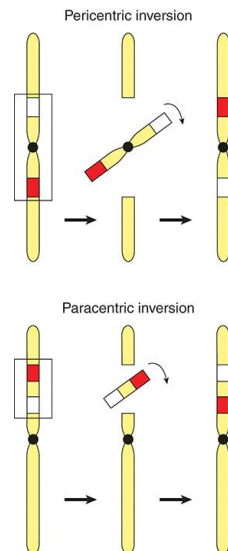


Figure 16-8. Pericentric inversion spans the centromere (dup/del offspring); paracentric does not (early pregnancy loss).

Ring chromosome

- Deletion at both ends → ends join. If only **telomeres** are lost → balanced carrier; more proximal loss → abnormal phenotype. **Ring X → Turner syndrome.**

5. Mosaicism

- ≥2 cytogenetically distinct cell lines **from one zygote**. Phenotype depends on which tissues (fetus, part of fetus, placenta) carry the abnormal line.
- Found in **~0.3% of amniotic fluid cultures**.
 - Abnormal cells in **one flask only** → probably **pseudomosaicism** (culture artifact).
 - **Multiple cultures** → true mosaicism more likely; confirmed in **60–70%**.

Confined placental mosaicism (CPM)

- Mosaicism in **1–2% of CVS specimens** → **offer amniocentesis**.
- In >1000 CVS mosaics: **true fetal mosaicism 13%**, **CPM 85%**, **uniparental disomy 2%**.
- If the chromosome has **imprinted genes** — **6, 7, 11, 14, 15** — consider **UPD testing**.
- Outcomes generally good, but ↑ **FGR and stillbirth**. One series: **preterm birth 45%**, **SGA 50%**.
- **⚠ CPM for trisomy 16** → poor prognosis (**as few as 1 in 3** normal outcomes).

Gonadal (germline) mosaicism

- One cell line in gametes, another in somatic cells → explains **apparently de novo** disease in children of normal parents.
- Spermatogonia divide throughout adulthood → new errors (paternal age >40).
- Explains the **~6% recurrence risk** after a child with a "new" mutation.

6. Modes of Inheritance

- **Monogenic (mendelian):** mutation at a single locus — autosomal dominant, autosomal recessive, X-linked, Y-linked.
- Other monogenic patterns: **mitochondrial, uniparental disomy, imprinting, trinucleotide repeat expansion.**

Table 16-3. Selected Monogenic (Mendelian) Disorders

Autosomal Dominant	Autosomal Recessive	X-Linked
Achondroplasia	α_1 -Antitrypsin deficiency	Androgen insensitivity syndrome
Acute intermittent porphyria	Congenital adrenal hyperplasia	Chronic granulomatous disease
Adult polycystic kidney disease	Cystic fibrosis	Color blindness
Antithrombin III deficiency	Gaucher disease	Fabry disease
BRCA1 and BRCA2 breast and/or ovarian cancer	Hemochromatosis	Fragile X syndrome
Ehlers-Danlos syndrome	Homocystinuria	Glucose-6-phosphate deficiency
Familial adenomatous polyposis	Phenylketonuria	Hemophilia A and B
Familial hypercholesterolemia	Sickle-cell anemia	Hypophosphatemic rickets
Hereditary hemorrhagic telangiectasia	Tay-Sachs disease	Muscular dystrophy — Duchenne and Becker
Hereditary spherocytosis	Thalassemia syndromes	Ocular albinism type 1 and 2
Huntington disease	Wilson disease	
Hypertrophic obstructive cardiomyopathy		
Long QT syndrome		
Marfan syndrome		
Myotonic dystrophy		
Neurofibromatosis type 1 and 2		
Tuberous sclerosis		
von Willebrand disease		

The printed table is a single column in three sections; laid out side by side here.

Phenotype vs genotype

- It is the **phenotype** that is dominant or recessive.
- Recessive carriers make some abnormal product but the normal co-gene directs the phenotype: sickle trait RBCs have **~30% HbS / 70% HbA** → do not sickle in vivo.

Heterogeneity

Type	Meaning	Example
Locus	Same phenotype from mutations at different loci (may follow different inheritance)	Retinitis pigmentosa (≥ 35 loci; AD, AR or X-linked)
Allelic	Different mutations in the same gene → variable severity	Cystic fibrosis: one gene (CFTR), >2000 mutations
Phenotypic	Different diseases from different mutations in the same gene	FGFR3 : achondroplasia and thanatophoric dysplasia

Autosomal dominant

- One copy determines the phenotype; **50% transmission** per conception.
- **Penetrance** = whether the gene is expressed at all (e.g., 80% penetrant). **Incomplete penetrance** → disease "skips" generations.
- **Expressivity** = range of severity in those who express it — e.g., **neurofibromatosis, tuberous sclerosis, adult polycystic kidney disease**.
- **Codominance**: both alleles expressed — **ABO blood groups, HbS + HbC** (Ch 59).

Advanced paternal age (>40)

- Spermatogonia divide **every 16 days** → ↑ single base-pair mutations → new **AD disorders** and **X-linked carrier states** ("paternal age effect disorders"):
 - **FGFR2** → craniosynostosis (**Apert, Crouzon, Pfeiffer**)
 - **FGFR3** → **achondroplasia**, thanatophoric dysplasia
 - **RET** → MEN syndromes
- **~2 extra mutations per year** of paternal age past 40.
- Absolute risk for any one disorder is low → **no specific screening or testing recommended**.

Autosomal recessive

- Disease only when **both copies** are abnormal. Carriers usually found after an affected child.
- **Recurrence 25%** per pregnancy: ¼ normal, ½ carriers, ¼ affected. **⅔ of unaffected siblings are carriers**.
- A carrier's risk depends on the partner: low unless **consanguinity** or an at-risk group.

Inborn errors of metabolism — absent enzyme → toxic intermediates → intellectual disability etc.

Phenylketonuria (PAH deficiency)

- AR; **PAH** converts phenylalanine → tyrosine. **>600** mutations.
- Untreated: progressive intellectual impairment, autism, seizures, motor deficits; **hypopigmentation** (phenylalanine inhibits tyrosine hydroxylase → ↓ melanin).
- Northern European carrier frequency **1 in 50**; disease up to **1 in 10,000** newborns. **Universal newborn screening**.
- Treatment: dietary phenylalanine restriction from early infancy + **phenylalanine-free amino-acid supplements**.
- **Pregnancy target: phenylalanine 2–6 mg/dL (120–360 µmol/L), from ≥3 months before conception and throughout pregnancy**.
- Phenylalanine is **actively transported to the fetus** → even a heterozygous (unaffected) fetus is damaged: miscarriage and **PKU embryopathy** (intellectual disability, microcephaly, seizures, growth impairment, cardiac anomalies).
- **Unrestricted diet**: intellectual disability **>90%**, microcephaly **>70%**, cardiac defects **up to 1 in 6**.
- Maternal PKU Collaborative Study (572 pregnancies): levels in range → ↓ anomalies and **normal childhood IQ**.
- **Sapropterin** (tetrahydrobiopterin, synthetic PAH cofactor): ↓ phenylalanine in **25–50%**; an option when diet fails.
- Preconceptional counseling + PKU-center care.

Consanguinity

- Union of **second cousins or closer**. **~10%** of the global population; 20–50% of marriages in some countries.
- Shared genes: 1st-degree **½**, 2nd-degree **¼**, 3rd-degree (first cousins) **⅙**.
- **First cousins: 2x risk** of congenital abnormalities; ↑ stillbirth; ↑ rare AR and multifactorial disorders.
- **SNP-array CMA can reveal consanguinity** → include in pretest counseling.
- **Incest** (1st-degree relatives): up to **40%** of offspring abnormal (older studies).

X-linked and Y-linked

- **Most X-linked diseases are recessive**: color blindness, **hemophilia A and B, Duchenne/Becker** muscular dystrophy.

- Affected male → **no affected sons** (sons get his Y); all daughters are carriers.
- Carrier woman → each **son 50% affected**, each **daughter 50% carrier**.
- **Skewed lyonization** → symptomatic carriers: **~10% of hemophilia A carriers have factor VIII <30%** (similar for factor IX in hemophilia B) → ↑ bleeding at delivery (risk exists even at higher levels).
- **DMD/BMD carriers** → **cardiomyopathy risk** → periodic cardiac/neuromuscular evaluation.
- **X-linked dominant** disorders mainly affect females (lethal in males): **vitamin D-resistant rickets, incontinentia pigmenti**. Exception: fragile X.
- **Y-linked** disorders are rare. **Yq deletions** → **severe spermatogenic defects**; Yp tip genes → meiotic pairing and fertility.

Mitochondrial inheritance

- Oocyte **~100,000 mitochondria**; sperm **~100** (destroyed after fertilization) → **exclusively maternal inheritance**. Males can be affected but **cannot transmit**.
- mtDNA: **16.5-kb circular** molecule, **37 genes** (oxidative phosphorylation, rRNAs, tRNAs).
- **Replicative segregation** → **homoplasmy** (all normal or all mutant) vs **heteroplasmy** (mixed). Degree of heteroplasmy in offspring **cannot be predicted** → counseling is difficult.
- **33** mitochondrial diseases with known molecular basis: **MERRF, Leber optic atrophy, Kearns-Sayre, Leigh syndrome**, mitochondrial myopathies/cardiomyopathies.

DNA triplet repeat expansion

- Repeats expand in transmission: during **male meiosis (Huntington)** or **female meiosis (fragile X)**.
- Clinical hallmark: **anticipation** — earlier onset or greater severity in each generation.

Table 16-4. Some Disorders Caused by DNA Triplet Repeat Expansion

Disorder
Dentatorubral-pallidoluysian atrophy
Fragile X syndrome
Friedreich ataxia
Huntington disease
Spinal and bulbar muscular atrophy
Myotonic dystrophies
Spinocerebellar ataxias

Fragile X syndrome

- **Most common inherited intellectual disability**: **~1 in 3600 males, 1 in 4000–6000 females**.
- **CGG** repeat at **Xq27.3**; full mutation → **FMR1 gene methylated and silenced**.
- Speech/language problems, ADHD, autism. Males: **IQ 35–45**. Narrow face, large jaw, **prominent ears**, connective tissue abnormalities, **macroorchidism** (postpubertal).

Category	CGG repeats
Unaffected	<45
Intermediate	45–54
Premutation	55–200
Full mutation	>200

- Full mutation is expressed in **all males** and many females (variable because of X-inactivation).
- Expansion to a full mutation almost always occurs through the **mother**.
- **Premutation mother** → **full-mutation child**: **≤5% with <70 repeats, >95% with 100–200 repeats**.

- Male premutation carrier: expansion extremely unlikely, but **all daughters carry the premutation**.
- Premutation prevalence: **~1 in 250** women without risk factors; **~1 in 90** with a family history of intellectual disability.
- Premutation health effects: **males** → **FXTAS** (tremor/ataxia, memory loss, dementia); **females** → **20% risk of fragile X-associated primary ovarian insufficiency (POI)**.
- **Carrier screening for:** family history of fragile X; unexplained intellectual disability, developmental delay or autism; **POI**.
- Prenatal diagnosis: amniocentesis or CVS (both give CGG repeat number); ⚠ **CVS may not accurately show FMR1 methylation status**.

Imprinting

- Expression depends on the **parent of origin** (epigenetic, genotype unchanged).
 - **Maternal imprinting:** maternal copy silent, paternal active. **Paternal imprinting:** the opposite.
- Imprint is **reset** in each generation (a female passes a paternally inherited gene with a maternal imprint).
- Explains diandric vs digynic triploidy phenotypes.

	Prader-Willi	Angelman
Region	15q11.2–q13	15q11.2–q13
Missing contribution	Paternal	Maternal (gene: ubiquitin protein ligase E3A)
Microdeletion	>70% (paternal chromosome 15)	~70% (maternal chromosome 15)
Other causes	Maternal UPD , or maternal imprinting with the paternal gene inactivated	Paternal imprinting defects 2–3% ; paternal UPD 2%
Features	Hyperphagia/obesity , short stature, small hands and feet, mild intellectual disability	Normal size; severe intellectual disability , absent speech , seizures, ataxia, jerky arm movements, inappropriate laughter

Memory hook: Prader-Willi = **P**aternal copy missing; Angelman = m**A**ternal copy missing.

Table 16-5. Some Disorders That Can Involve Imprinting

Disorder	Chromosomal Region	Parental Origin
Angelman	15q11.2–q13	Maternal
Beckwith-Wiedemann	11p15.5	Paternal
Myoclonus-dystonia	7q21	Maternal
Prader-Willi	15q11.2–q13	Paternal
Pseudohypoparathyroidism	20q13.2	Variable
Russell-Silver syndrome	7p11.2	Maternal

Uniparental disomy (UPD)

- **Both members of a pair from one parent.** Usually harmless, **except chromosomes 6, 7, 11, 14, 15** (imprinted genes).
- Commonest mechanism: **trisomic rescue** — loss of one homologue from a trisomic conceptus → UPD in **~1/3**.
- **Isodisomy** = two identical copies from one parent → e.g., **cystic fibrosis with only one carrier parent**.

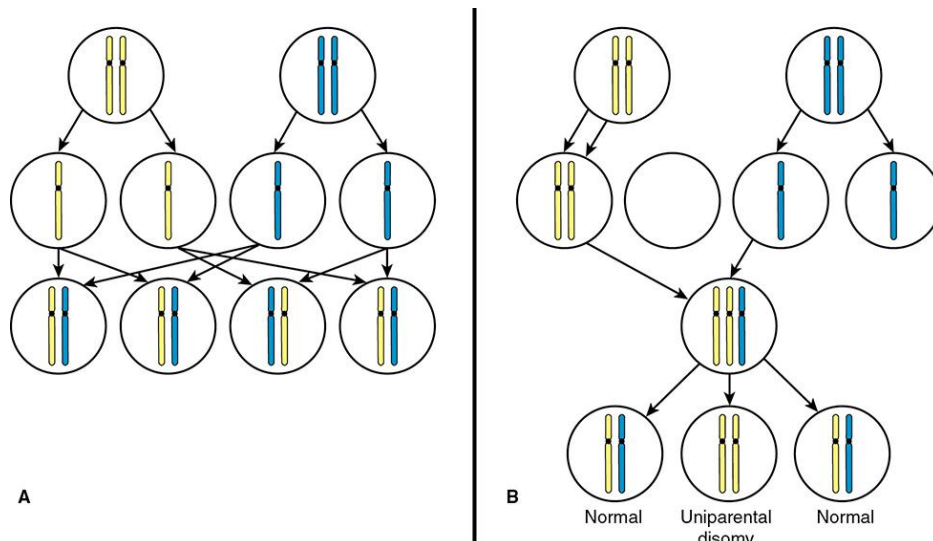


Figure 16-9. Trisomic rescue: after nondisjunction, loss of one of three homologues gives uniparental disomy in about a third.

Multifactorial inheritance

- Multiple genes + environment (**polygenic** = several genes). Examples: **NTDs, cardiac defects, facial clefts**, diabetes, atherosclerosis, height, head size.
- Familial but non-mendelian. **After one affected child: empiric recurrence 3–5%**; falls exponentially with more distant relationships.

Table 16-6. Characteristics of Multifactorial Diseases

Characteristic	Details
There is a genetic contribution	No mendelian pattern of inheritance; no evidence of single-gene disorder
Nongenetic factors are also involved in disease causation	Lack of penetrance despite predisposing genotype; monozygotic twins may be discordant
Familial aggregation may occur	Relatives are more likely to have disease-predisposing alleles
Expression more common among close relatives	Greater concordance in monozygotic than dizygotic twins; becomes less common in less closely related relatives — fewer predisposing alleles

- **Continuously variable traits:** normal distribution; **>2 SD** from the mean = abnormal; offspring are less extreme (**regression to the mean**).
- **Threshold traits:** all-or-none, appear only when liability exceeds a threshold.
 - If the **less-often-affected sex** is affected, relatives' risk is higher. **Pyloric stenosis ~4× more common in males** → an affected female's children/siblings have >3–5% risk, and **her sons and brothers have the highest risk**.
 - **Severity** raises recurrence: **bilateral cleft lip + palate ~8%** vs **unilateral cleft lip ~4%**.

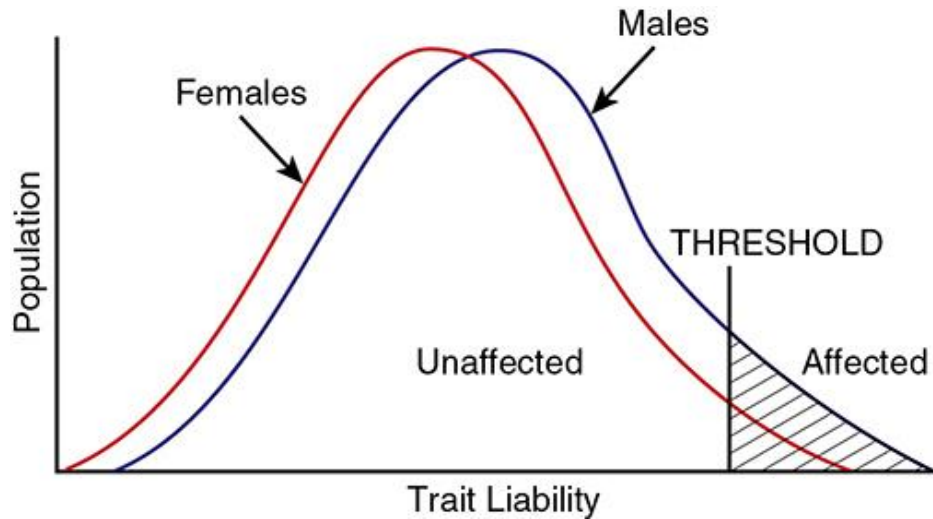


Figure 16-10. Threshold trait with male predilection: at the same threshold more males than females are affected.

Cardiac defects

- **Most common birth defects: 8 per 1000 births.**
- Offspring risk: **5–6% if mother affected, 2–3% if father affected.**
- Left-sided lesions (**HLHS, coarctation, bicuspid aortic valve**) → recurrence **4–6× higher**. See Table 52-4.

Neural-tube defects

- Risk factors: family history, **hyperthermia, hyperglycemia**, teratogens, **obesity**, ethnicity, fetal sex.
- Most NTDs occur without maternal folate deficiency (complex gene–nutrient interactions).
- **Prior affected child: recurrence 3–5%, ↓ ≥70% with 4 mg/d periconceptional folic acid.**
- Folic acid also appears to help with pregestational diabetes, antiepileptic drugs, fever in embryogenesis and obesity.

7. Genetic Tests

- **All pregnant women** should be offered **aneuploidy screening** (serum or cfDNA) **and diagnostic testing**.
- Carrier screening for **cystic fibrosis and spinal muscular atrophy** routinely; ethnic-specific, panethnic and expanded panels are all acceptable (Ch 17).
- Diagnosis: **CMA, karyotype, FISH** on amniotic fluid or chorionic villi; **PCR** for specific known single-gene diseases. WGS/WES not routine.

Cytogenetic analysis (karyotyping)

- Detects **aneuploidy, polyploidy** and balanced/unbalanced rearrangements **≥5–10 Mb. Accuracy >99%.**
- Needs **dividing cells** (arrested in metaphase). **Giemsa → G-bands**. Metaphase banding **450–550 bands**; prophase **~850**.
- Turnaround:

Sample	Time
Amniotic fluid	7–10 days
Fetal blood	36–48 h (rarely needed)
Postmortem skin fibroblasts	2–3 wk (largely replaced by CMA)

Fluorescence in situ hybridization (FISH)

- **Rapid (1–2 days)** targeted test for **21, 18, 13** (and X, Y) → used when the result may change management. Can confirm 22q11.2 deletion (CMA preferred).

- Fluorescent probes hybridize to complementary DNA; **number of signals = number of copies**. Looks **only at the region probed**, not the whole complement.
- ⚠️ ACOG: **don't act on FISH alone** — need supporting data (abnormal screen, sonographic finding, or karyotype/CMA confirmation).

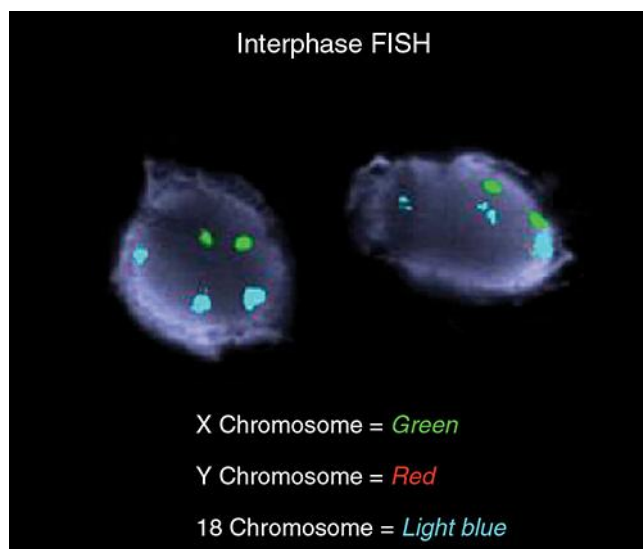


Figure 16-12. Interphase FISH: three light-blue (18) signals, two green (X), no red (Y) = female fetus with trisomy 18.

Chromosomal microarray analysis (CMA)

- **100× more sensitive than karyotype**; detects deletions/duplications as small as **50–100 kb**.
- Turnaround: **3–7 days** direct (uncultured cells); **10–14 days** if culture needed. **No dividing cells needed**.
- **CGH platform**: fetal DNA and control DNA labeled with different dyes, hybridized to oligonucleotides on a chip; signal intensities compared.
- **SNP platform**: chip of known SNPs; signal intensity = copy number.

Detects	CGH array	SNP array
Aneuploidy, unbalanced translocations, microdeletions/duplications	✓	✓
Triploidy	✗	✓
Absence of heterozygosity (UPD, consanguinity)	✗	✓
Balanced rearrangements	✗	✗

- ⚠️ Because CMA misses balanced rearrangements, **recurrent pregnancy loss → karyotype first-line**.
- **Variants of uncertain significance ~1%** of prenatal specimens (source of parental distress).

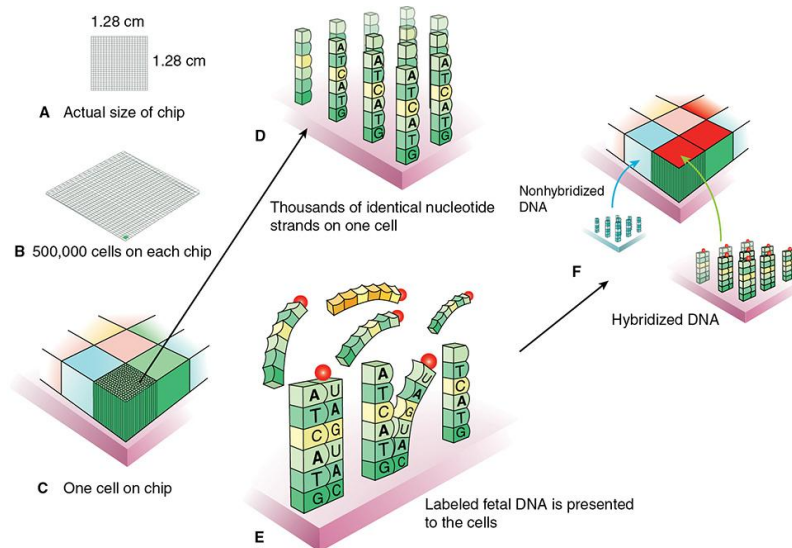


Figure 16-13. CMA chip: ~500,000 cells, each holding thousands of identical oligonucleotides; labeled fetal DNA binds complementary sequences and glows.

Clinical applications

- High-risk aneuploidy screen → offer **karyotype or FISH + karyotype**; CMA should be available.
- Normal karyotype → CMA finds relevant CNVs in **~6.5% with fetal anomalies** and **1–2% without**.
- **Fetal structural anomaly → CMA is the first-tier test** (ACOG, SMFM).
 - Exception: an anomaly strongly suggesting a specific aneuploidy (**endocardial cushion defect → T21**; **aloobar holoprosencephaly → T13**) → karyotype or FISH first.
- CMA may reveal unmanifested AD disease in a parent or **nonpaternity**.
- **Stillbirth**: CMA beats karyotype (no dividing cells). SCRNI: **~6%** extra diagnoses when karyotype uninformative. Meta-analysis: pathogenic CNV in **6%** with anomalies and **3%** without; CMA gave a result **15% more often** than karyotype.

Whole genome (WGS) and whole exome (WES) sequencing

- WGS = entire genome; **WES = coding regions (~1.5%)**. Used postnatally for suspected syndromes/intellectual disability.
- Fetus with anomalies + normal karyotype + normal CMA: **WES diagnoses ~10%**; highest yield with **cardiac, skeletal or multisystem** anomalies.
- Limitations: long turnaround, cost, false-positive/negative results, VUS, incidental actionable findings in parents (trio sequencing).
- **Not recommended for routine prenatal use.**

8. Fetal DNA in the Maternal Circulation

- **Intact fetal cells**: only **2–6 cells/mL**; persist for **decades** → **microchimerism** (implicated in scleroderma, SLE, Hashimoto thyroiditis). Limited diagnostic use (low numbers, persistence into later pregnancies, hard to tell apart from maternal cells).

Cell-free DNA (cfDNA; NIPS)

- Fragments from maternal cells + **apoptotic placental trophoblast** — so "fetal" is a misnomer.
- Reliably detected **after 9–10 wk**. **Fetal fraction ≈ 10%** of total cfDNA. **Cleared from maternal blood within minutes** (unlike intact fetal cells).
- Current clinical uses: **aneuploidy screening** and **Rh D genotyping**.

- Microdeletion panels and genome-wide screening (**>7 Mb**) are offered commercially but **not recommended routinely**.
- Research: single-gene disorders (skeletal dysplasias, hemoglobinopathies, CF, CAH, hemophilia, muscular/ myotonic dystrophy, SMA).

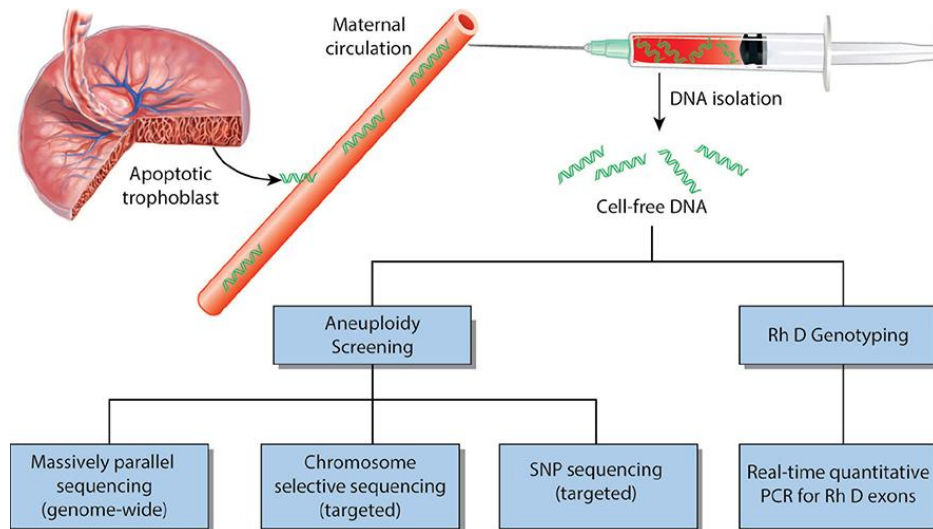


Figure 16-14. cfDNA from apoptotic trophoblast: aneuploidy screening (massively parallel, chromosome-selective or SNP sequencing) and Rh D genotyping (real-time PCR).

Aneuploidy screening

- Platforms: **massively parallel (whole-genome) sequencing**, chromosome-selective/targeted sequencing, **SNP analysis**. A T21 fetus → excess proportion of chromosome 21 fragments.
- Performance (meta-analysis, mostly high-risk):

Condition	Sensitivity
Trisomy 21	99.7%
Trisomy 18	98%
Trisomy 13	99%
Specificity (each)	≥99.9% — cumulative false-positive rate <1%

- **Fetal sex**: sensitivity **95% at 7–12 wk**, **99% after 20 wk** (useful for X-linked disease, CAH).
- **⚠ Screening only — confirm with diagnostic testing before any irreversible intervention.**
- **No result in 2–4%** (assay failure, high variance, **low fetal fraction**) → these pregnancies have **↑ aneuploidy risk**.
- A result may reflect **CPM**, a **vanished aneuploid co-twin**, **maternal mosaicism** or rarely **occult maternal cancer**.

Table 16-7. Selected Etiologies Creating Inaccurate Cell-free DNA Results

Abnormality not detected — false negative	Normal fetus but abnormal screen — false positive
Low fetal fraction: fetal chromosomal abnormality; multifetal gestation; maternal obesity; low-molecular-weight heparin	Confined placental mosaicism (normal fetus, aneuploid placenta)
Confined placental mosaicism (normal placenta, aneuploid fetus)	Multifetal gestation with loss of aneuploid fetus
	Maternal aneuploidy (e.g., 47,XXX)
	Maternal mosaicism (e.g., 46,XX/45,X) or fetal mosaicism
	Maternal cancer (lymphoma, breast, colon, leukemia, others)
	Vitamin B ₁₂ deficiency

Rh D genotyping

- **~40%** of fetuses of Rh D-negative (white) women are also Rh D-negative → avoid unnecessary **anti-D**, and in alloimmunization avoid needless **MCA Doppler/amniocentesis**.
- **Real-time PCR** targeting RHD **exons 4, 5, 7 and 10**. Routine in **Denmark, Finland, Netherlands**.
- At **24–27 wk** (>35,000 women): **false-negative 0.03%**, false-positive **<1%**. UK: false-negative higher in the 1st trimester; added alloimmunization risk **<1 per million births**.

Quick-Recall: Inheritance Patterns and Recurrence Risks

Situation	Risk / key fact
Prior autosomal trisomy	~1% recurrence (or age risk if higher)
Prior triploidy (past 1st trimester)	1–1.5%
Autosomal dominant parent	50% per pregnancy
Both parents AR carriers	25% affected; 2/3 of unaffected sibs are carriers
X-linked recessive carrier mother	Sons 50% affected , daughters 50% carriers ; affected father → no affected sons, all daughters carriers
Mitochondrial	Only maternal transmission
"New" dominant mutation (gonadal mosaicism)	~6%
Multifactorial, one affected child	3–5% (bilateral CL/P 8%, unilateral CL 4%)
NTD recurrence	3–5% , ↓ ≥70% with 4 mg/d folic acid
Congenital heart disease in parent	Mother 5–6% , father 2–3% ; left-sided lesions 4–6× higher
Reciprocal translocation carrier	5–30% (after abnormal child) vs ~5% (other ascertainment)
Robertsonian carrier	~15% maternal , ~2% paternal ; homologous → 100% unbalanced gametes
Pericentric inversion	5–10% (after abnormal child) vs 1–3% ; inv(9) normal variant
Paracentric inversion	Extremely low
Fragile X premutation mother	≤5% (<70 repeats) → >95% (100–200 repeats)

Test	Resolution / turnaround	Misses
Karyotype	5–10 Mb ; 7–10 days (amnio)	Microdeletions
FISH	Targeted; 1–2 days	Everything not probed
CMA	50–100 kb ; 3–7 days direct	Balanced rearrangements ; triploidy (CGH)
WES	Coding 1.5% of genome	Not routine prenatally
cfDNA	Screening; after 9–10 wk	Not diagnostic; 2–4% no-call

Key Numbers to Memorize

- Chromosomal abnormalities: **>50%** of 1st-trimester losses; **0.4%** of pregnancies; T21 **>50%**, T18 **15%**, T13 **5%** of cases
- Oocyte aneuploidy **10–20%** vs sperm **3–4%**
- Down syndrome **1:450 pregnancies / 1:740 live births**; 95% nondisjunction (75% meiosis I), 3–4% robertsonian; cardiac **50%**; life expectancy **55–60 yr**
- T18 **1:2000**; heart defects **>90%**; median survival **8 days**, 5-yr **12%**
- T13 **1:5000**; holoprosencephaly **2%**; median survival **5 days**; maternal hypertension **≥25%**
- Turner **1:2500** girls; 98% lost early; paternal X missing **80%**; left heart lesions **30–50%**
- Klinefelter **1:700** males; XXX and XYY **1:1000** each
- 22q11.2 deletion **1:3000–6000**, **>90% de novo**; conotruncal heart **75%**
- Robertsonian **1:1000**; reciprocal **1:600**; inv(9) **1%** of people
- Mosaicism **0.3%** of amnio, **1–2%** of CVS (CPM 85%, true 13%, UPD 2%)
- Imprinted chromosomes **6, 7, 11, 14, 15**
- PKU target **2–6 mg/dL**, from **3 months** preconception
- Fragile X: normal **<45**, intermediate **45–54**, premutation **55–200**, full **>200**; premutation **1:250** women; POI **20%**
- CMA: **50–100 kb**, CNV in **6.5%** of anomalous fetuses with normal karyotype; VUS **~1%**; WES **~10%** extra yield
- cfDNA: fetal fraction **~10%**; T21 sensitivity **99.7%**; no-call **2–4%**; Rh D false-negative **0.03%**